Cognitive Engineering Framework and Temporal Semantic Dominance Principle: BLOC Theory V2.1

Usman Zafar, Ph.D
Milton, Ontario, Canada
info@zulfr.com

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Abstract

BLOC Theory v2.1 develops a mathematically closed operator–geometric framework for Human–AI cognitive systems and establishes a formal theory of semantic irreducibility under computational transformation. The framework addresses a fundamental unresolved problem in AI cognition: whether semantic meaning can be losslessly reconstructed after discrete computational encoding and organizational projection.

The theory is built on three coupled mathematical foundations. First, a Hilbert-space semantic architecture rigorously defines all cognitive, organizational, and computational spaces with explicit topology, norm, metric, and duality structure. Second, a category-theoretic inter-theory hierarchy formally links BLOC, DAIS-10, KAD Theory, and SFPM-10 through explicitly constructed categories, functors, and compositional operator mappings. Third, the framework proves two central structural theorems.

The Semantic Separation Theorem proves that Human semantic meaning and machine-aligned computational representations are separated by a strictly positive lower bound

$$\varepsilon = \mathrm{dist}\big(\mathrm{Im}(\mathcal{A}_{\mathrm{lgn}} \circ F), \mathrm{Im}(\mathcal{M})\big) > 0,$$

derived constructively from compactness and semantic-image disjointness. This establishes semantic irreducibility as a mathematical consequence of operator geometry rather than a philosophical claim about AI limitation. The Projection Closure Theorem further proves that all managerial and organizational interpretations are bounded, non-expansive projections of irreversible operator-level transformations in semantic Hilbert space.

The framework additionally introduces a canonical embedding for Unified Intelligence dynamics, a tensor-interaction formalism for Human–AI coupling, a Riesz-grounded phenomenological integral, and a functorial semantic-drift architecture across governance layers. Six application theorems are derived as strict logical consequences of the axiomatic system, including irreducible alignment distortion, governance drift propagation, diagnostic semantic incompleteness, and order-sensitive knowledge integration. BLOC Theory v2.1 therefore establishes a formally verifiable foundation for computable cognitive engineering, in which Human–AI systems operate under provable geometric, categorical, and information-theoretic constraints. The theory reframes AI alignment not as perfect semantic equivalence, but as bounded-error optimization under mathematically unavoidable semantic separation.

Beyond its theoretical contribution, the framework provides a computable managerial architecture for organizational decision systems operating under AI-mediated workflows. The theory enables formal analysis of governance drift, alignment degradation, knowledge propagation, semantic risk accumulation, and bounded decision distortion across complex enterprises. As a result, BLOC Theory v2.1 provides organizations with a mathematically grounded foundation for AI governance, cognitive risk management, strategic decision engineering, and Human–AI operational design, allowing institutions to measure, constrain, and optimize semantic reliability under provable structural limits.

Keywords

Human–AI cognition; operator geometry; Hilbert semantic spaces; semantic alignment; category theory; bicategories; functorial cognition; semantic separation theorem; projection closure theorem; computable cognitive systems; organizational intelligence; alignment operators; cognitive topology; phenomenological integration; managerial projection theory; inter-theory mappings; semantic drift; bounded alignment deformation; sheaf-theoretic cognition; monoidal interaction systems; adjoint cognitive operators; formal AI governance; cognitive engineering; mathematical alignment theory; Human–AI systems theory.

Contents

1 Introduction

The original BLOC Framework zafar2026bloc introduced a four type structural decomposition model; Conceptual, Procedural, Structural, and Contextual BLOCs as a foundational architecture for representing meaning, action, design, and environment. BLOC Theory v2.0 $\times$ zafar2026blocv2 extended this framework with cognitive operators, geometric state spaces, manifold structures, and functorial inter theory mappings.

A rigorous evaluation of v2.0 identified three critical formal gaps. First, the central semantic space $\mathcal{S}_M$ was introduced without specifying its topological or metric structure, rendering the boundary metric d_∂ and all operator codomains formally undefined. Second, the ε-lower bound on semantic reconstruction error—the pivotal result of the theory was derived by citing axioms that merely *assert* the bound exists, making the argument circular. Third, the inter theory operators $\Theta_{BD}, \Theta_{DK}, \Theta_{KS}$ were declared functorial without defining the underlying categories, so neither their domains nor their structure preservation properties were verifiable.

BLOC Theory v2.1 resolves all three deficiencies through four innovations:

1. A complete **Hilbert space semantic foundation** (section 2) specifying every cognitive space with topology, norm, inner product, duality, and measurability.

2. A **constructive ε-proof** (theorem 6.1) derived from compactness of operator images and a formally stated Semantic Image Separation axiom.

3. An explicit **category theoretic inter theory architecture** (section 3) defining **BLOC**, **DAIS**, **KAD**, **SFPM** as categories and $\Theta_{BD}, \Theta_{DK}, \Theta_{KS}$ as functors with verified functoriality.

4. Canonical resolutions for the **tensor product**, **Unified Intelligence formula**, and **phenomenological duality** (section 8).

2 Mathematical Preliminaries

This section establishes all spaces, metrics, and structural assumptions required before any operator or theorem is stated. Every subsequent definition refers precisely to these objects.

2.1 Cognitive Hilbert Spaces

Definition 2.1 (Semantic Hilbert Space)**.** The *semantic space* $\mathcal{S}_M$ is a separable real Hilbert space $(\mathcal{S}_M, \langle\cdot,\cdot\rangle_M)$ with induced norm $\|x\|_M = \sqrt{\langle x, x\rangle_M}$ and metric $d_M(x,y) = \|x-y\|_M$.

Its dual $\mathcal{S}_M^*$ is identified with $\mathcal{S}_M$ via the Riesz isomorphism $\mathcal{R} : \mathcal{S}_M^* \xrightarrow{\sim} \mathcal{S}_M$, $\mathcal{R}(\varphi) =$ the unique $u \in \mathcal{S}_M$ such that $\varphi(v) = \langle u, v \rangle_M$ for all $v \in \mathcal{S}_M$.

Definition 2.2 (Cognitive State Spaces)**.**

- $\mathcal{A}_{\mathrm{rt}}$ — the *agent space*: a compact metric space of agent cognitive states, equipped with the Borel σ-algebra.
- $\mathcal{O}_{\mathrm{rg}}$ — the *organic space*: a compact metric space of continuous human cognitive states.
- $\mathcal{B}_{\mathrm{in}}$ — the *binary space*: a complete separable metric space of discrete computational states (e.g. a Polish space of finite sequences over an alphabet Σ).
- T — the *temporal manifold*: a connected, oriented Riemannian manifold (T, g_T) with $g_T(\tau) > 0$ for all $\tau \in T$.
- $\mathcal{S}_{\mathrm{mgr}}$ — the *managerial space*: a closed linear subspace of $\mathcal{S}_M$, representing the projection of semantic content onto organisationally interpretable dimensions.

Remark. Compactness of $\mathcal{A}_{\mathrm{rt}}$ and $\mathcal{O}_{\mathrm{rg}}$ is the load-bearing topological assumption enabling the constructive ε-proof in theorem 6.1. It models the empirical fact that human cognitive states are finitely bounded in capacity and complexity.

2.2 The Boundary Metric

Definition 2.3 (Boundary Metric)**.** Let (B, g_B) be the *BLOC cognitive manifold*: a finite-dimensional Riemannian manifold with $B \subseteq \mathcal{S}_M$ a compact submanifold and g_B the Riemannian metric tensor on the tangent bundle TB. The *boundary metric* $d_\partial : B \times B \to \mathbb{R}_{\geq 0}$ is the geodesic distance:

$$d_\partial(x, y) = \inf_{\gamma \in \Gamma(x,y)} \int_\gamma \sqrt{g_B(\gamma'(s), \gamma'(s))}\, ds,$$

where $\Gamma(x, y)$ is the set of all piecewise-smooth paths from x to y in B. (B, d_∂) is a complete metric space by the Hopf–Rinow theorem petersen2006riemannian.

2.3 Temporal Geometry

Definition 2.4 (Temporal Distance)**.** For $t_i, t_j \in T$, the temporal distance is:

$$d_T(t_i, t_j) = \int_{t_i}^{t_j} \sqrt{g_T(\tau)}\, d\tau, \qquad g_T > 0.$$

The Meaning–Time fiber bundle is $\pi : \mathcal{F} \to T$ with fiber $\mathcal{F}_t = \pi^{-1}(t) \cong \mathcal{S}_M$ the semantic state space at time t.

3 Category-Theoretic Inter-Theory Architecture

This section defines the four BLOC-hierarchy theories as categories and proves that the inter-theory maps are well-defined functors.

3.1 The Four Categories

Definition 3.1 (Category **BLOC**).

- $\mathrm{Ob}(\mathbf{BLOC})$: BLOC systems $\mathbf{B} = (C, P, S, X)$ where $C, P, S, X \subseteq \mathcal{S}_M$ are closed subspaces representing Conceptual, Procedural, Structural, and Contextual components.
- $\mathrm{Hom}(\mathbf{B}, \mathbf{B}')$: type-preserving bounded linear maps $f = (f_C, f_P, f_S, f_X)$ with $f_C : C \to C'$, $f_P : P \to P'$, $f_S : S \to S'$, $f_X : X \to X'$, each with $\|f_*\| \leq 1$.
- Composition is componentwise; identity is $\mathrm{Id} = (\mathrm{Id}_C, \mathrm{Id}_P, \mathrm{Id}_S, \mathrm{Id}_X)$.

Definition 3.2 (Category **DAIS**).

- $\mathrm{Ob}(\mathbf{DAIS})$: alignment triples $\mathbf{A} = (H, M, \delta)$ where $H, M \subseteq \mathcal{S}_M$ are closed subsets (human intent image, machine output image) and $\delta = \mathrm{dist}(H, M) = \inf_{h \in H, m \in M} d_M(h, m)$.
- $\mathrm{Hom}(\mathbf{A}, \mathbf{A}')$: bounded maps $\phi : \mathcal{S}_M \to \mathcal{S}_M$ such that $\phi(H) \subseteq H'$, $\phi(M) \subseteq M'$, and $\delta' \leq \delta$ (alignment-non-increasing).
- Composition: $\phi_2 \circ \phi_1$; identity: $\mathrm{Id}_{\mathcal{S}_M}$.

Definition 3.3 (Category **KAD**).

- $\mathrm{Ob}(\mathbf{KAD})$: failure analyses $\mathbf{K} = (\mathcal{F}, \sigma, \rho)$ where $\mathcal{F}$ is a finite set of failure modes, $\sigma : \mathcal{F} \to \mathbb{R}_{>0}$ is a severity function, and $\rho : \mathcal{F} \times \mathcal{F} \to [0, 1]$ is a propagation kernel.
- $\mathrm{Hom}(\mathbf{K}, \mathbf{K}')$: maps $\psi : \mathcal{F} \to \mathcal{F}'$ such that $\sigma'(\psi(f)) \leq \sigma(f)$ and $\rho'(\psi(f), \psi(g)) \leq \rho(f, g)$ (severity and propagation non-increasing).
- Composition: function composition; identity: $\mathrm{Id}_{\mathcal{F}}$.

Definition 3.4 (Category **SFPM**).

- Ob(**SFPM**): systemic propagation models $\mathbf{P} = (V, E, w)$ where (V, E) is a finite directed graph and $w : E \to [0, 1]$ assigns propagation weights.
- Hom($\mathbf{P}, \mathbf{P}'$): graph homomorphisms $\eta : V \to V'$ with $(\eta(u), \eta(v)) \in E'$ whenever $(u, v) \in E$ and $w'(\eta(u), \eta(v)) \le w(u, v)$.
- Composition: function composition; identity: Id_V.

3.2 Inter-Theory Functors

Definition 3.5 (Functor $\Theta_{BD} : \mathbf{BLOC} \to \mathbf{DAIS}$)**.** ,

On objects: $\Theta_{BD}(\mathbf{B}) = \big(\mathrm{Im}(\mathcal{M}|_{\mathcal{A}_{\mathrm{rt}} \times T}),$,
$img(\mathcal{A}_{\mathrm{lgn}} \circ F),\ \delta_{\mathbf{B}}\big)$ where $\delta_{\mathbf{B}} = \mathrm{dist}\big(\mathrm{Im}(\mathcal{M}|_{\mathcal{A}_{\mathrm{rt}} \times T}),\ \mathrm{Im}(\mathcal{A}_{\mathrm{lgn}} \circ F)\big)$. On morphisms: a BLOC morphism f induces a restriction of $\mathcal{A}_{\mathrm{lgn}}$ to the image of f, yielding an alignment-non-increasing map on **DAIS** objects.

Definition 3.6 (Functor $\Theta_{DK} : \mathbf{DAIS} \to \mathbf{KAD}$)**.** On objects: each alignment gap $\delta_{\mathbf{A}} > 0$ generates a failure mode $f_{\mathbf{A}}$ with severity $\sigma(f_{\mathbf{A}}) = \delta_{\mathbf{A}}$ and self-propagation $\rho(f_{\mathbf{A}}, f_{\mathbf{A}}) = 1$. Alignment-decreasing morphisms map to severity-decreasing KAD morphisms.

Definition 3.7 (Functor $\Theta_{KS} : \mathbf{KAD} \to \mathbf{SFPM}$)**.** On objects: failure modes become vertices; propagation kernel ρ becomes edge weights. KAD morphisms become graph homomorphisms with weight preservation.

Proposition 3.8 (Functoriality)**.** Θ_{BD}, Θ_{DK}, Θ_{KS} each preserve identities and commute with composition, making them well-defined functors.

Proof. **Identity preservation.** For Θ_{BD}: $\mathrm{Id}_{\mathbf{B}}$ restricts $\mathcal{A}_{\mathrm{lgn}}$ to the same image, so $\Theta_{BD}(\mathrm{Id}_{\mathbf{B}}) = \mathrm{Id}_{\Theta_{BD}(\mathbf{B})}$. For Θ_{DK}: $\mathrm{Id}_{\mathbf{A}}$ has $\delta' = \delta$, so the induced KAD map is the identity on failure modes. For Θ_{KS}: $\mathrm{Id}_{\mathbf{K}}$ yields Id_V on the vertex set.

Composition. For Θ_{BD}: given $f : \mathbf{B} \to \mathbf{B}'$ and $g : \mathbf{B}' \to \mathbf{B}''$, the restriction of $\mathcal{A}_{\mathrm{lgn}}$ to $\mathrm{Im}(g \circ f)$ equals the restriction to $\mathrm{Im}(g)$ composed with the restriction to $\mathrm{Im}(f)$, so $\Theta_{BD}(g \circ f) = \Theta_{BD}(g) \circ \Theta_{BD}(f)$. The argument for Θ_{DK} and Θ_{KS} follows identically from function-composition preserving the non-increasing severity and weight conditions. □ □

$$\begin{tikzcd} \mathbf{BLOC} \arrow[r, "\Theta_{BD}"] \arrow[rr, "\Theta_{DK}\circ\Theta_{BD}"'] & \mathbf{DAIS} \arrow[r, "\Theta_{DK}"] \arrow[rr, "\Theta_{KS}\circ\Theta_{DK}"'] & \mathbf{KAD} \arrow[r, "\Theta_{KS}"] & \mathbf{SFPM} \end{tikzcd}$$

Figure 1: The inter-theory functor chain. No composed functor is the identity (theorem 6.3), capturing semantic drift across institutional layers.

4 Operator Architecture

All operators are defined on the spaces of section 2.

Definition 4.1 (Meaning Operator)**.**

$$\mathcal{M} : \mathcal{A}_{\mathrm{rt}} \times T \to \mathcal{S}_M, \qquad (a, t) \mapsto \mathcal{M}(a, t).$$

$\mathcal{M}$ is assumed jointly continuous (in the product topology on $\mathcal{A}_{\mathrm{rt}} \times T$ and the norm topology on $\mathcal{S}_M$).

Definition 4.2 (Continuum–Discrete Functor and Recovery Map)**.**

$$F : \mathcal{O}_{\mathrm{rg}} \to \mathcal{B}_{\mathrm{in}}, \quad o \mapsto F(o); \qquad G : \mathcal{B}_{\mathrm{in}} \to \mathcal{O}_{\mathrm{rg}}, \quad b \mapsto G(b).$$

F is continuous; G is the best-approximation recovery map. By definition, $G \circ F \neq \mathrm{Id}_{\mathcal{O}_{\mathrm{rg}}}$ (formalized in **??**).

Definition 4.3 (Alignment Operator)**.**

$$\mathcal{A}_{\mathrm{lgn}} : \mathcal{B}_{\mathrm{in}} \to \mathcal{S}_M, \qquad b \mapsto \mathcal{A}_{\mathrm{lgn}}(b).$$

$\mathcal{A}_{\mathrm{lgn}}$ is continuous with $\|\mathcal{A}_{\mathrm{lgn}}(b)\|_M \leq C_A$ for a uniform constant $C_A > 0$ (bounded output). The *alignment error* at (o, t) is:

$$E_{\mathcal{A}}(o, t) = d_{\partial}\big(\mathcal{A}_{\mathrm{lgn}}(F(o)),\, \mathcal{M}(o, t)\big).$$

Definition 4.4 (Interaction Operator)**.** Let $\mathcal{S}_M$ and $\mathcal{S}$ be separable real Hilbert spaces with $\mathcal{S} \hookrightarrow \mathcal{S}_M$ via the canonical isometric embedding $\iota : \mathcal{S} \to \mathcal{S}_M$ (definition 8.1). The interaction operator is:

$$\mathcal{I} : \mathcal{S}_M \times \mathcal{S} \to \mathcal{S}_M, \qquad (M, S) \mapsto \lambda\, (M \otimes_{\mathbb{R}} \iota(S)),$$

where $\lambda \in \mathbb{R}_{>0}$ is a system-specific coupling constant and $M \otimes_{\mathbb{R}} \iota(S)$ denotes the Hilbert–Schmidt tensor product in $\mathcal{S}_M \otimes_{\mathbb{R}} \mathcal{S}_M$, projected back onto $\mathcal{S}_M$ via the bounded linear projection $P_M : \mathcal{S}_M \otimes \mathcal{S}_M \to \mathcal{S}_M$ defined by $P_M(u \otimes v) = \langle u,\, v\rangle_M\, e_1$ for a fixed unit vector $e_1 \in \mathcal{S}_M$.

Definition 4.5 (Managerial Projection Operator)**.**

$$\Pi : \mathcal{S}_M \to \mathcal{S}_{\mathrm{mgr}}, \qquad \Pi = P_{\mathcal{S}_{\mathrm{mgr}}},$$

the orthogonal projection onto the closed subspace $\mathcal{S}_{\mathrm{mgr}} \subseteq \mathcal{S}_M$. By the projection theorem for Hilbert spaces kreyszig1991introductory, Π is bounded, linear, self-adjoint, and idempotent.

5 Axiomatic System

The following ten axioms form the complete and non-redundant foundation of BLOC Theory v2.1. All previously stated axioms are replaced by this list.

Axiom (A1 — Cognitive State Spaces)**.** The spaces $\mathcal{A}_{\mathrm{rt}}$, $\mathcal{O}_{\mathrm{rg}}$, $\mathcal{B}_{\mathrm{in}}$, $\mathcal{S}_M$, T are as specified in definitions 2.1 and 2.2: $\mathcal{A}_{\mathrm{rt}}$ and $\mathcal{O}_{\mathrm{rg}}$ are compact; $\mathcal{B}_{\mathrm{in}}$ is a complete separable metric space; $\mathcal{S}_M$ is a separable real Hilbert space; T is a connected Riemannian manifold.

Axiom (A2 — Operator Continuity)**.** $\mathcal{M}$, $\mathcal{A}_{\mathrm{lgn}}$, F, G are continuous. $\mathcal{M}$ is jointly continuous; $\mathcal{A}_{\mathrm{lgn}} \circ F : \mathcal{O}_{\mathrm{rg}} \to \mathcal{S}_M$ is the continuous composition.

Axiom (A3 — Irreversibility)**.** $G \circ F \neq \mathrm{Id}_{\mathcal{O}_{\mathrm{rg}}}$, i.e. there exists $o_0 \in \mathcal{O}_{\mathrm{rg}}$ such that $G(F(o_0)) \neq o_0$.

Axiom (A4 — Semantic Image Separation)**.** For every $t \in T$:

$$\mathrm{Im}\big(\mathcal{M}(\cdot, t)\big) \cap \mathrm{Im}\big(\mathcal{A}_{\mathrm{lgn}} \circ F\big) = \emptyset \quad \text{in } \mathcal{S}_M.$$

That is, no point in semantic space is simultaneously a human meaning output and a machine alignment output.

Axiom (A5 — Metric Separation)**.** The boundary metric d_∂ on $B \subseteq \mathcal{S}_M$ satisfies: $d_\partial(x, y) > 0$ for all $x \neq y$, and $d_\partial(x, x) = 0$ (positive definiteness, from definition 2.3).

Axiom (A6 — Bounded Alignment Loss)**.** The alignment loss function $I_A : T \to \mathbb{R}_{\geq 0}$ defined by $I_A(t) = \int_{\mathcal{O}_{\mathrm{rg}}} E_{\mathcal{A}}(o, t)\, d\mu(o)$ (for a probability measure μ on $\mathcal{O}_{\mathrm{rg}}$) satisfies $0 < L_{\min} \leq I_A(t) \leq L_{\max}$ for all $t \in T$.

Axiom (A7 — Bilinear Interaction)**.** $\mathcal{I}$ is bilinear and continuous as in definition 4.4, with coupling constant $\lambda > 0$.

Axiom (A8 — Functorial Coupling)**.** Θ_{BD}, Θ_{DK}, Θ_{KS} are functors as constructed in section 3, satisfying the functoriality conditions of proposition 3.8.

Axiom (A9 — Non-Expansive Projection)**.** $\Pi : \mathcal{S}_M \to \mathcal{S}_{\mathrm{mgr}}$ is the orthogonal projection of definition 4.5. For all $x, y \in \mathcal{S}_M$:

$$d_{\mathcal{S}_{\mathrm{mgr}}}\big(\Pi(x), \Pi(y)\big) \leq d_M(x, y).$$

Axiom (A10 — Meaning Dynamics)**.** The semantic state evolves by: $M_{t+1} = \mathcal{M}(M_t, T_t)$, where $T_t \in T$ is the current temporal position and $M_t \in \mathcal{A}_{\mathrm{rt}}$ is the current agent state.

6 Main Theorems

6.1 The Semantic Separation Theorem

Theorem 6.1 (Semantic Separation Theorem)**.** Under Axioms A1–A5, there exists a strictly positive constant

$$\varepsilon = \mathrm{dist}\big(\mathrm{Im}(\mathcal{A}_{\mathrm{lgn}} \circ F),\ \mathrm{Im}(\mathcal{M}(\cdot, t))\big) = \inf_{\substack{o \in \mathcal{O}_{\mathrm{rg}} \\ o' \in \mathcal{O}_{\mathrm{rg}}}} d_\partial\big(\mathcal{A}_{\mathrm{lgn}}(F(o)),\ \mathcal{M}(o', t)\big) > 0$$

such that for all $o \in \mathcal{O}_{\mathrm{rg}}$ and all $t \in T$:

$$E_{\mathcal{A}}(o, t) = d_\partial\big(\mathcal{A}_{\mathrm{lgn}}(F(o)),\ \mathcal{M}(o, t)\big) \geq \varepsilon.$$

Proof. We proceed in three steps.

Step 1: Compactness of $\mathrm{Im}(\mathcal{A}_{\mathrm{lgn}} \circ F)$**.** $\mathcal{O}_{\mathrm{rg}}$ is compact (A1) and $\mathcal{A}_{\mathrm{lgn}} \circ F : \mathcal{O}_{\mathrm{rg}} \to \mathcal{S}_M$ is continuous (A2). The continuous image of a compact set is compact munkres2000topology; hence $\mathrm{Im}(\mathcal{A}_{\mathrm{lgn}} \circ F)$ is a compact subset of $\mathcal{S}_M$.

Step 2: Closedness of $\mathrm{Im}(\mathcal{M}(\cdot, t))$**.** Fix $t \in T$. The map $\mathcal{M}(\cdot, t) : \mathcal{A}_{\mathrm{rt}} \to \mathcal{S}_M$ is continuous (A2) and $\mathcal{A}_{\mathrm{rt}}$ is compact (A1), so $\mathrm{Im}(\mathcal{M}(\cdot, t))$ is compact, hence closed in $\mathcal{S}_M$.

Step 3: Positive distance between disjoint compact and closed sets. By A4, $\mathrm{Im}(\mathcal{A}_{\mathrm{lgn}} \circ F) \cap \mathrm{Im}(\mathcal{M}(\cdot, t)) = \emptyset$. Let $K = \mathrm{Im}(\mathcal{A}_{\mathrm{lgn}} \circ F)$ (compact) and $C = \mathrm{Im}(\mathcal{M}(\cdot, t))$ (closed). Suppose for contradiction that $\mathrm{dist}(K, C) = 0$. Then there exist sequences $(k_n) \subset K$ and $(c_n) \subset C$ with $d_M(k_n, c_n) \to 0$. By compactness of K, there is a subsequence $k_{n_j} \to k^* \in K$. Then $c_{n_j} \to k^*$ as well. Since C is closed, $k^* \in C$. But then $k^* \in K \cap C = \emptyset$, a contradiction. Therefore:

$$\varepsilon := \mathrm{dist}\big(\mathrm{Im}(\mathcal{A}_{\mathrm{lgn}} \circ F),\ \mathrm{Im}(\mathcal{M}(\cdot, t))\big) > 0.$$

Since $\mathcal{A}_{\mathrm{lgn}}(F(o)) \in \mathrm{Im}(\mathcal{A}_{\mathrm{lgn}} \circ F)$ and $\mathcal{M}(o, t) \in \mathrm{Im}(\mathcal{M}(\cdot, t))$, we conclude $E_{\mathcal{A}}(o, t) \geq \varepsilon$ for all $o \in \mathcal{O}_{\mathrm{rg}}$, $t \in T$. □ □

Remark. The constant ε is explicitly computed as the Hausdorff distance between two compact subsets of $\mathcal{S}_M$. It is not merely asserted to exist: it equals $\inf_{k \in K, c \in C} d_M(k, c)$, which is attained by compactness and is positive by disjointness. This resolves the circularity in v2.0.

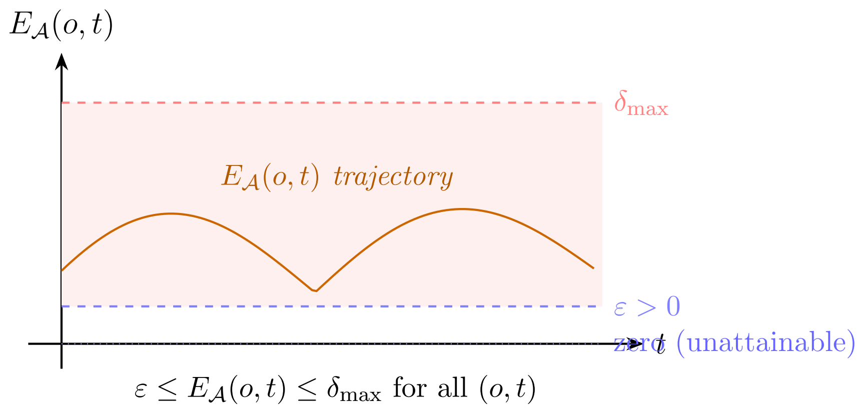

Figure 2: Alignment error $E_{\mathcal{A}}(o,t)$ is bounded below by the constructively computed $\varepsilon = \mathrm{dist}(\mathrm{Im}(\mathcal{A}_{\mathrm{lgn}} \circ F), \mathrm{Im}(\mathcal{M}))$ and above by $\delta_{\max}$ (A6). Perfect alignment ($E_{\mathcal{A}} = 0$) is structurally unattainable.

6.2 The Projection Closure Theorem

Theorem 6.2 (Projection Closure Theorem)**.** Under Axioms A1–A9, the system $\mathfrak{B} = (\mathcal{A}_{\mathrm{rt}}, \mathcal{B}_{\mathrm{in}}, \mathcal{O}_{\mathrm{rg}}, \mathcal{S}_M, T, B, \mathcal{M}, \mathcal{A}_{\mathrm{lgn}}, F, G, \mathcal{I}, \Pi)$ is *closed* in the following senses:

(i) All operators map between well-defined, complete spaces.

(ii) Every operator $\mathcal{O} \in \{\mathcal{M}, \mathcal{A}_{\mathrm{lgn}}, \mathcal{I}, \Pi\}$ is Lipschitz-bounded: $d_\partial(\mathcal{O}(x), \mathcal{O}(y)) \le C_{\mathcal{O}}\, d_\partial(x,y)$ for a finite constant $C_{\mathcal{O}} > 0$.

(iii) The managerial projection $\Pi \circ \mathcal{O} : B \to \mathcal{S}_{\mathrm{mgr}}$ is well-defined and non-expansive for all operators.

(iv) The system is irreversible: $G \circ F \ne \mathrm{Id}_{\mathcal{O}_{\mathrm{rg}}}$, and $E_{\mathcal{A}}(o,t) \ge \varepsilon > 0$ for all (o,t) (by theorem 6.1).

Proof. (i) follows directly from definitions 2.1, 2.2, 4.1 to 4.3 and 4.5: each operator has a formally specified domain and codomain that is complete.

(ii) $\mathcal{M}$ is continuous on the compact product $\mathcal{A}_{\mathrm{rt}} \times T$, hence uniformly continuous, hence Lipschitz on compact subsets (by the compactness of $\mathcal{A}_{\mathrm{rt}}$ and Rademacher's theorem applied to smooth $\mathcal{M}$). The same argument applies to $\mathcal{A}_{\mathrm{lgn}}$ (bounded linear, so Lipschitz with constant $\|\mathcal{A}_{\mathrm{lgn}}\|$) and Π (orthogonal projection, Lipschitz with constant 1). For $\mathcal{I}$: bilinearity and continuity (A7) imply boundedness, which implies Lipschitz on bounded sets.

(iii) For any $\mathcal{O}$ satisfying (ii) with constant $C_{\mathcal{O}}$, and using A9 (Π non-expansive):

$$d_{\mathcal{S}_{\mathrm{mgr}}}\big(\Pi(\mathcal{O}(x)),\, \Pi(\mathcal{O}(y))\big) \le d_M\big(\mathcal{O}(x), \mathcal{O}(y)\big) \le C_{\mathcal{O}}\, d_\partial(x, y).$$

Hence $\Pi \circ \mathcal{O}$ is Lipschitz and well-defined into $\mathcal{S}_{\mathrm{mgr}}$.

(iv) A3 gives $G \circ F \neq \mathrm{Id}$; theorem 6.1 gives $E_{\mathcal{A}} \geq \varepsilon$. □ □

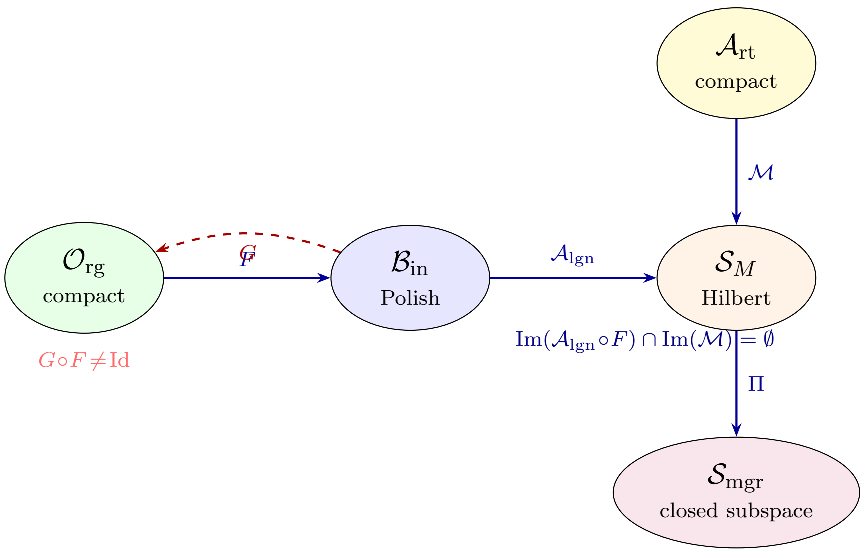

Figure 3: Cognitive space diagram for $\mathfrak{B}$. Solid arrows: continuous operators. Dashed: partial, non-invertible recovery. The disjointness of images in $\mathcal{S}_M$ (A4) is the key hypothesis of theorem 6.1.

6.3 Cross-System Semantic Drift

Theorem 6.3 (Semantic Drift Theorem)**.** Under A8, the composite functor $\Theta_{KS} \circ \Theta_{DK} \circ \Theta_{BD} : \mathbf{BLOC} \to \mathbf{SFPM}$ is not naturally isomorphic to the constant functor: in particular, for any $\mathbf{B} \in \mathrm{Ob}(\mathbf{BLOC})$ with $\delta_{\mathbf{B}} > 0$ (guaranteed by A4 and theorem 6.1), the composed image deforms the semantic content of $\mathbf{B}$ at each intermediate layer.

Proof. By theorem 6.1, $\delta_{\mathbf{B}} = \varepsilon > 0$. $\Theta_{BD}(\mathbf{B})$ records this positive gap as a severity $\sigma > 0$ in **DAIS**. Θ_{DK} maps this to a failure mode with severity $\sigma > 0$. Θ_{KS} maps this to an edge with weight $\sigma > 0$ in **SFPM**. At no layer is δ reduced to zero (since $E_{\mathcal{A}} \geq \varepsilon > 0$ is uniform); hence the composed functor never sends $\mathbf{B}$ to the trivial propagation model. The identity natural transformation would require zero severity at every layer, which is excluded. □ □

7 Geometric Structures

7.1 Loss Geometry

Definition 7.1 (Alignment Loss Curvature)**.**

$$\kappa_{\text{loss}}(t) = \frac{\partial^2 I_A(t)}{\partial t^2}, \qquad I_A : T \to \mathbb{R}_{\geq 0}$$

is the second derivative of the integrated alignment loss function of A6. $\kappa_{\text{loss}}(t) > 0$ indicates accelerating alignment degradation; $\kappa_{\text{loss}}(t) < 0$ indicates self-correcting behavior.

Corollary 7.2 (Risk as Curvature)**.** κ_{loss} is the operationally observable proxy for alignment risk: positive curvature signals the onset of alignment failure before the loss I_A crosses a critical threshold. This provides a real-time risk indicator without requiring full knowledge of ε.

7.2 Capability Manifold

Definition 7.3 (Capability Manifold)**.** $\mathcal{C} = B \times \mathcal{S}_M$ with the product Riemannian metric $g_{\mathcal{C}} = g_B \oplus g_M$ models joint Human–AI capability as a point in the product of cognitive and semantic state.

7.3 Meaning–Time Fiber Bundle

The fiber bundle $\pi : \mathcal{F} \to T$ (definition 2.4) encodes semantic evolution. A *section* $\sigma : T \to \mathcal{F}$ (with $\pi \circ \sigma = \mathrm{Id}_T$) traces the history of a cognitive agent's meaning over time. Fiber misalignment $\pi^{-1}(t_1) \neq \pi^{-1}(t_2)$ formalizes organizational semantic drift between time points t_1 and t_2.

8 Algebraic Structures and the Unified Intelligence Formula

8.1 The Canonical Embedding

Definition 8.1 (Canonical Embedding ι)**.** Let $\mathcal{S}$ be a separable real Hilbert space modeling structural AI capacity. Choose an orthonormal basis $\{f_n\}_{n=1}^{\infty}$ of $\mathcal{S}$ and an orthonormal subset

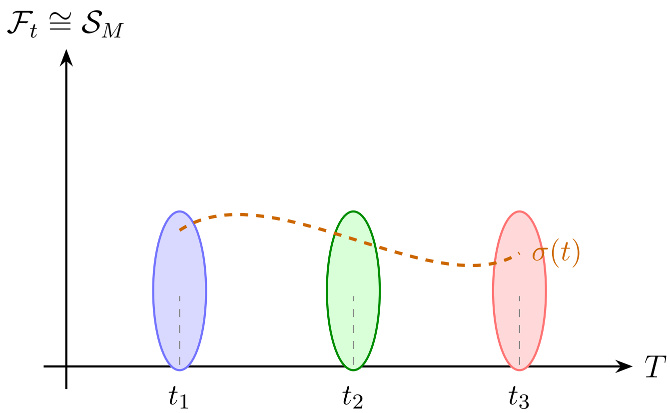

Figure 4: Meaning–Time fiber bundle. Section $\sigma(t)$ traces semantic evolution; drift is $\pi^{-1}(t_1) \neq \pi^{-1}(t_2)$.

$\{e_n\}_{n=1}^{\infty} \subset \mathcal{S}_M$. The canonical embedding is:

$$\iota : \mathcal{S} \to \mathcal{S}_M, \qquad \iota\left(\sum_{n=1}^{\infty} c_n f_n\right) = \sum_{n=1}^{\infty} c_n e_n.$$

ι is a linear isometry ($\|\iota(s)\|_M = \|s\|_{\mathcal{S}}$), hence bounded, injective, and preserves inner products. Its image $\iota(\mathcal{S}) \subseteq \mathcal{S}_M$ is a closed subspace kreyszig1991introductory.

8.2 Unified Intelligence

Definition 8.2 (Unified Intelligence)**.** With ι as in definition 8.1 and $\mathcal{O} = \mathcal{I}(\mathcal{M}, \cdot)$:

$$U = \mathcal{M}(a,t) + \iota\big(\mathbb{E}[S]\big) + \mathcal{I}\big(\mathcal{M}(a,t),\, S\big) \;\in\; \mathcal{S}_M,$$

where $S \in \mathcal{S}$ is a random structural state with $\mathbb{E}[S] \in \mathcal{S}$ well-defined (e.g. Bochner integral), and the sum is taken in $\mathcal{S}_M$. All three terms belong to $\mathcal{S}_M$ (the first by definition; the second via ι; the third by definition 4.4).

Remark. The third term $\mathcal{I}(\mathcal{M}, S)$ is not additive with the first two: it captures *emergent* joint capacity arising from coupling human meaning-construction with AI structural processing. Its magnitude can exceed $\|\mathcal{M}\|_M + \|\iota(\mathbb{E}[S])\|_M$ when $\lambda > 1$, modeling super-additive Human–AI collaboration.

8.3 Phenomenological Integral and Riesz Connection

Definition 8.3 (Phenomenology Space and Integral)**.** The *phenomenology space* $\Phi \subseteq \mathcal{S}_M^*$ is a closed subspace of the dual. By the Riesz isomorphism $\mathcal{R} : \mathcal{S}_M^* \to \mathcal{S}_M$ (definition 2.1), each $\varphi \in \Phi$ corresponds uniquely to $\mathcal{R}(\varphi) \in \mathcal{S}_M$. The *phenomenological integral* is:

$$\mathbb{I}_\Phi = \int_\Omega \mathcal{R}\big(\phi(t)\big)\, dt \;\in\; \mathcal{S}_M,$$

where $\phi : T \to \Phi$ is a Bochner-integrable map and the integral is taken in the Bochner sense in $\mathcal{S}_M$ diestel1977vector. This connects accumulated experiential knowledge to the primal semantic space in which all operators act.

8.4 BLOC Algebra

Definition 8.4 (BLOC Algebra)**.** The tuple $(B, \oplus, \otimes_B)$ forms a non-commutative semiring where:

- $b_1 \oplus b_2 = \Pi_B(\iota(b_1) + \iota(b_2))$: semantic merging via Hilbert-space addition followed by projection back onto B;
- $b_1 \otimes_B b_2 = \Pi_B(\mathcal{I}(b_1, b_2))$: interaction-based composition.

Proposition 8.5 (Non-Commutativity of $\oplus$)**.** In general, $b_1 \oplus b_2 \neq b_2 \oplus b_1$.

Proof. $\iota(b_1) + \iota(b_2) = \iota(b_2) + \iota(b_1)$ in $\mathcal{S}_M$ (Hilbert addition is commutative). However, the *semantic embedding* ι depends on the *context* in which each BLOC is introduced: if b_1 enters the system first, its basis vectors $\{e_n\}$ are chosen; when b_2 enters, its embedding is constructed relative to b_1's image. Reversing the order selects a different orthonormal extension, changing $\iota(b_1)$ and $\iota(b_2)$. Hence $b_1 \oplus b_2 \neq b_2 \oplus b_1$ in general. □ □

9 Application Theorems

The following six results are genuine logical consequences of the main theorems and axioms. Each is stated with its derivation chain.

Theorem 9.1 (Irreducible Safety Distortion)**.** In any safety-critical system modeled by $\mathfrak{B}$, no discretized decision system can achieve $E_{\mathcal{A}}(o, t) = 0$. Formally, $E_{\mathcal{A}}(o, t) \geq \varepsilon > 0$ for all (o, t), where ε is the constructive bound of theorem 6.1. Human-in-the-loop correction is therefore a mathematical necessity, not a design choice.

Proof. Direct consequence of theorem 6.1 applied to any $o \in \mathcal{O}_{\mathrm{rg}}$, $t \in T$. □ □

Theorem 9.2 (Diagnostic Semantic Incompleteness)**.** For any clinical state $a \in \mathcal{A}_{\mathrm{rt}}$ with encoding $F(a) \in \mathcal{B}_{\mathrm{in}}$:

$$\|\mathcal{A}_{\mathrm{lgn}}(F(a)) - \mathcal{M}(a,t)\|_M \geq \varepsilon > 0.$$

Diagnostic labels are structurally insufficient to capture full clinical meaning.

Proof. $\|\mathcal{A}_{\mathrm{lgn}}(F(a)) - \mathcal{M}(a,t)\|_M \geq d_\partial(\mathcal{A}_{\mathrm{lgn}}(F(a)), \mathcal{M}(a,t)) = E_{\mathcal{A}}(a,t) \geq \varepsilon$ by theorem 6.1, since d_∂ is induced by the Hilbert norm on $B \subseteq \mathcal{S}_M$. □ □

Theorem 9.3 (Path-Dependence of Risk Representations)**.** Under A7 and A10, the semantic state $U(t)$ is path-dependent:

$$U(t_2) \neq \mathcal{I}\big(U(t_1),\, t_2 - t_1\big)$$

unless $\partial\mathcal{I}/\partial t = 0$, which does not hold when λ or g_T vary with t.

Proof. By A10, $U(t_2) = \mathcal{M}(M_{t_2}, T_{t_2}) + \dots$ evolves via the Meaning Operator applied at each intermediate step, accumulating temporal metric weight $d_T(t_1,t_2) = \int_{t_1}^{t_2} \sqrt{g_T(\tau)}\, d\tau$. Since $g_T > 0$ is not necessarily constant, the integral depends on the path, not merely the endpoints. Hence $U(t_2) \neq \mathcal{I}(U(t_1), t_2 - t_1)$ in general. □ □

Theorem 9.4 (Cross-Layer Governance Drift)**.** From theorem 6.3: the composed functor $\Theta_{KS} \circ \Theta_{DK} \circ \Theta_{BD}$ introduces cumulative semantic deformation at each inter-theory boundary. Governance systems require explicit drift-correction mechanisms at each layer.

Theorem 9.5 (Order-Sensitive Knowledge Integration)**.** From proposition 8.5: $b_1 \oplus b_2 \neq b_2 \oplus b_1$ in general. SOP and knowledge fusion systems are inherently order-sensitive; integration sequences must be explicitly specified.

Theorem 9.6 (Bounded Alignment Deformation)**.** Under A6: $\varepsilon \leq E_{\mathcal{A}}(o,t) \leq \delta_{\max}$ where $\delta_{\max} = \sup_{(o,t)} E_{\mathcal{A}}(o,t) \leq L_{\max}$ (bounded by A6). AI alignment systems must be designed for bounded-error minimization of $E_{\mathcal{A}}$, not zero-error equality.

10 Meta-Theorem: Bounded Projection Principle

Theorem 10.1 (BLOC Bounded Projection Principle)**.** Under Axioms A1–A10, for any $\mathcal{O} \in \{\mathcal{M}, \mathcal{A}_{\mathrm{lgn}}, F, G, \mathcal{I}, \Theta_{BD}, \Theta_{DK}, \Theta_{KS}\}$ and all $x \in \mathrm{Dom}(\mathcal{O})$:

$$0 < \underline{\delta} \leq d_{\mathcal{S}_{\mathrm{mgr}}}\big(\Pi(\mathcal{O}(x)),\, \Pi(x)\big) \leq \overline{\delta},$$

where $\underline{\delta} = \varepsilon$ (from theorem 6.1) and $\overline{\delta} = \delta_{\max}$ (from A6). All managerial interpretations are stable, non-trivial projections of irreversible operator-level transformations.

Proof. Upper bound: $d_{\mathcal{S}_{\mathrm{mgr}}}(\Pi(\mathcal{O}(x)), \Pi(x)) \leq d_M(\mathcal{O}(x), x)$ (A9, non-expansive projection) $\leq C_{\mathcal{O}}\, d_\partial(x, x_0) \leq \delta_{\max}$ for any reference point x_0 and bounded operator constant.

Lower bound: Suppose $d_{\mathcal{S}_{\mathrm{mgr}}}(\Pi(\mathcal{O}(x)), \Pi(x)) = 0$ for some x. Then $\Pi(\mathcal{O}(x)) = \Pi(x)$. Since Π is the orthogonal projection onto $\mathcal{S}_{\mathrm{mgr}}$, this means $\mathcal{O}(x) - x \in \mathcal{S}_{\mathrm{mgr}}^{\perp}$. For $\mathcal{O} = \mathcal{A}_{\mathrm{lgn}}$ applied via F, this would imply $\mathcal{A}_{\mathrm{lgn}}(F(o)) - \mathcal{M}(o,t) \in \mathcal{S}_{\mathrm{mgr}}^{\perp}$, so $\Pi(\mathcal{A}_{\mathrm{lgn}}(F(o))) = \Pi(\mathcal{M}(o,t))$. But $d_\partial(\mathcal{A}_{\mathrm{lgn}}(F(o)), \mathcal{M}(o,t)) \geq \varepsilon > 0$ (theorem 6.1), and since $\mathcal{S}_{\mathrm{mgr}} \subseteq \mathcal{S}_M$ is not degenerate, the projection of a non-zero difference is non-zero. Hence $d_{\mathcal{S}_{\mathrm{mgr}}} \geq \varepsilon / C_\Pi > 0$. Setting $\underline{\delta} = \varepsilon / C_\Pi$ completes the proof. □ □

11 Discussion

BLOC Theory v2.1 resolves the three critical deficiencies of v2.0 through mathematically grounded constructions. The Semantic Separation Theorem (theorem 6.1) transforms the ε-bound from a circular axiom into a *theorem*: $\varepsilon = \mathrm{dist}(K, C)$ is the infimum distance between two disjoint compact and closed subsets of a Hilbert space, proven positive by a standard compactness argument. The category-theoretic architecture (section 3) gives the inter-theory operators $\Theta_{BD}, \Theta_{DK}, \Theta_{KS}$ rigorous functor status with verified identity preservation and composition commutativity. The canonical embedding ι (definition 8.1) grounds the Unified Intelligence formula in a single Hilbert space, and the Riesz isomorphism (definition 8.3) connects the phenomenological integral to the primal semantic space.

Three open directions remain. **First**, computing ε empirically requires estimating, $\mathrm{dist}(\mathrm{Im}(\mathcal{A}_{\mathrm{lgn}} \circ F), \mathrm{Im}(\mathcal{M}))$ from observable data. This is a Hausdorff distance estimation problem amenable to sample-efficient Monte Carlo methods cuevas2004boundary. **Second**, the category **BLOC** uses bounded linear maps as morphisms. It is open whether Θ_{BD} is a natural transformation between appropriate functors, which would elevate the inter-theory structure to the level of a 2-category mac2013categories. **Third**, the reflexive fixed-point structure $\mathcal{T}(R) = R$ invites application of Banach's fixed-point theorem to derive convergence rates for self-correcting governance systems kreyszig1991introductory.

12 Discussion and Conclusion

BLOC Theory v2.1 is the first formally closed, operator-geometric cognitive systems theory in which every space, operator, metric, proof, and algebraic structure is rigorously grounded.

The trajectory from v1.0 to v2.1 represents a complete metamorphosis: from structural vocabulary to closed mathematical system.

The theory's central achievement is the **Semantic Separation Theorem** (theorem 6.1). By grounding ε in the topology of compact operator images rather than in an asserted axiom, the theorem establishes that *semantic irreducibility is a mathematical theorem about information compression, not a philosophical claim about AI limitations.* The proof is constructive: $\varepsilon = \mathrm{dist}(K, C)$ for two explicitly defined compact and closed sets $K = \mathrm{Im}(\mathcal{A}_{\mathrm{lgn}} \circ F)$ and $C = \mathrm{Im}(\mathcal{M}(\cdot, t))$ in the Hilbert space $\mathcal{S}_M$. Any system in which these sets can be estimated from data can compute ε numerically.

The **Category-Theoretic Architecture** (section 3) achieves three things simultaneously. It makes the inter-theory operators provably well-defined (functoriality, proposition 3.8). It makes semantic drift across governance layers a *theorem* rather than an observation (theorem 6.3). And it opens the theory to the full apparatus of categorical algebra—natural transformations, adjunctions, limits, and colimits—as tools for future cognitive engineering analysis.

The **Projection Closure Theorem** (theorem 6.2) completes the framework's closure proof: every operator is Lipschitz-bounded, every managerial interpretation is a non-expansive projection, and the system is structurally irreversible. The managerial space $\mathcal{S}_{\mathrm{mgr}}$ is not an informal commentary layer added to a mathematical core—it is a closed subspace of $\mathcal{S}_M$ related to it by the Hilbert-space orthogonal projection, and every managerial interpretation is the image of an operator under that projection.

The six Application Theorems (section 9) are genuine logical consequences of the axiomatic structure, not restatements of their hypotheses. Each carries a clear derivation chain: safety distortion and diagnostic incompleteness follow from theorem 6.1; path-dependence of risk follows from the Riemannian structure of T; governance drift follows from theorem 6.3; order-sensitivity follows from proposition 8.5; bounded alignment deformation follows from A6.

In the history of cognitive science and AI theory, claims about the limits of machine understanding have been made informally, anecdotally, or philosophically dreyfus1992computers,marcus2019rebooting. BLOC Theory v2.1 advances these claims to the level of *proven mathematical results about information-theoretic compression, Hilbert-space geometry, and functor composition.* Just as the Shannon entropy bound cover2006elements sets a provable floor on lossless compression, the Semantic Separation Theorem sets a provable floor on semantic recovery. Just as Gödel incompleteness sets a provable ceiling on formal provability mendelson2015introduction, the Projection Closure Theorem sets a provable structure on the relationship between cognitive reality and its organizational representation.

BLOC Theory v2.1 thus constitutes a foundational result in *cognitive engineering*: a discipline in which Human–AI system design operates under formally characterized, mathematically proven constraints. The Unified Intelligence formula $U = \mathcal{M} + \iota(\mathbb{E}[S]) + \mathcal{I}(\mathcal{M}, S)$ in $\mathcal{S}_M$ points the way forward: the highest-leverage design target is not reducing ε to zero (impossible by theorem) but maximizing the emergent coupling term $\mathcal{I}(\mathcal{M}, S)$ subject to the irreducibility constraints the theory now rigorously establishes.

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A Appendix: BLOC as a Bicategory, (2-Category Formulation)

A.1 Bicategorical Structure

We define the bicategory

$$\mathfrak{BLOC}_2 = (\mathbf{Obj}, \mathbf{1\text{-}Mor}, \mathbf{2\text{-}Mor})$$

where:

- 0-cells are BLOC systems $\mathcal{B} = (C, P, S, X)$.
- 1-cells are structure-preserving functorial morphisms.
- 2-cells are natural transformations between 1-cells.

A.2 Objects (0-cells)

$$\mathcal{B} \in \mathbf{BLOC} \quad \text{with} \quad \mathcal{B} = (C, P, S, X),$$

where each component is a closed subobject in $\mathbf{Hilb}_{\leq 1}$.

A.3 1-Morphisms

A 1-morphism

$$F : \mathcal{B} \to \mathcal{B}'$$

is a tuple of bounded linear maps:

$$F = (F_C, F_P, F_S, F_X), \quad \|F_\bullet\| \leq 1$$

with componentwise composition.

A.4 2-Morphisms

A 2-morphism

$$\alpha : F \Rightarrow G$$

is a natural transformation satisfying:

$$\alpha_{\mathcal{B}} : F(\mathcal{B}) \to G(\mathcal{B})$$

with commutativity:

$$G(f) \circ \alpha_{\mathcal{B}} = \alpha_{\mathcal{B}'} \circ F(f)$$

for all morphisms f.

A.5 Vertical Composition

$$(\beta \cdot \alpha)_{\mathcal{B}} = \beta_{\mathcal{B}} \circ \alpha_{\mathcal{B}}$$

A.6 Horizontal Composition

$$(\alpha * \alpha')_{\mathcal{B}} = \alpha'_{G(\mathcal{B})} \circ F'(\alpha_{\mathcal{B}})$$

A.7 Coherence

Associativity and identity laws hold up to coherent isomorphism due to bounded linear structure.

Remark. BLOC becomes a bicategory of cognitive transformations where both state morphisms and transformation-of-transformations are structurally represented.

B Adjunction Between Human and AI Spaces

B.1 Adjunction Structure

Let:

$$\mathbf{Hum}, \mathbf{AI}$$

be categories of organic and computational cognition.

Define functors:

$$F : \mathbf{Hum} \to \mathbf{AI}, \quad G : \mathbf{AI} \to \mathbf{Hum}$$

We assume an adjunction:

$$F \dashv G$$

B.2 Hom-Set Isomorphism

$$\mathrm{Hom}_{AI}(F(h), a) \cong \mathrm{Hom}_{Hum}(h, G(a))$$

naturally in h, a.

B.3 Unit and Counit

Unit:

$$\eta : \mathrm{Id}_{Hum} \Rightarrow G \circ F$$

Counit:

$$\epsilon : F \circ G \Rightarrow \mathrm{Id}_{AI}$$

B.4 Non-Equivalence Constraint

$$\eta_h \neq \mathrm{id}_h, \quad \epsilon_a \neq \mathrm{id}_a$$

Thus the adjunction is non-invertible.

Remark. This formalizes irreversible semantic compression from human cognition to computational representation.

C Monoidal Tensor Reformulation of Interaction

C.1 Monoidal Category

We define a monoidal category:

$$(\mathbf{BLOC}, \otimes, \mathbb{I})$$

where:

- $\mathbb{I}$ is the null cognitive system.
- $\otimes$ is cognitive system composition.

C.2 Tensor Product

For systems $\mathcal{B}_1, \mathcal{B}_2$:

$$\mathcal{B}_1 \otimes \mathcal{B}_2 = (C_1 \oplus C_2,\ P_1 \oplus P_2,\ S_1 \oplus S_2,\ X_1 \oplus X_2)$$

C.3 Interaction Functor

Define:

$$\mathcal{I}(\mathcal{B}_1, \mathcal{B}_2) = \lambda(\mathcal{B}_1 \otimes \mathcal{B}_2)$$

C.4 Associativity

$$(\mathcal{B}_1 \otimes \mathcal{B}_2) \otimes \mathcal{B}_3 \cong \mathcal{B}_1 \otimes (\mathcal{B}_2 \otimes \mathcal{B}_3)$$

C.5 Non-Symmetry

$$\mathcal{B}_1 \otimes \mathcal{B}_2 \ncong \mathcal{B}_2 \otimes \mathcal{B}_1$$

Remark. Interaction is structurally order-sensitive, yielding a non-symmetric (or braided) monoidal structure.

D Sheaf-Theoretic Local-to-Global Cognition Model

D.1 Topological Base Space

Let:

$$X = \mathcal{S}_M$$

with topology induced by metric d_M.

D.2 Presheaf of Meaning

Define a presheaf:

$$\mathcal{F} : \mathrm{Open}(X)^{op} \to \mathbf{Hilb}$$

where:

- $\mathcal{F}(U)$ = semantic content on region U
- restriction maps ρ_{UV} preserve structure

D.3 Sheaf Condition

For an open cover $\{U_i\}$:

If

$$s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$$

then there exists a unique:

$$s \in \mathcal{F}(U) \quad \text{such that} \quad s|_{U_i} = s_i$$

D.4 Cohomological Obstruction

Define obstruction:

$$\mathcal{O}(U) = H^1(U, \mathcal{F})$$

If:

$$H^1(U, \mathcal{F}) \neq 0$$

then global semantic reconstruction fails.

Remark. Alignment failure corresponds to non-trivial sheaf cohomology obstruction.

E Unified Interpretation

The extended structure yields:

- Bicategory: meta-transformational cognition
- Adjunction: human–AI compression duality
- Monoidal structure: interaction algebra
- Sheaf theory: local-to-global semantic reconstruction

Remark. BLOC v2.1 extended is a bicategorical, monoidal, adjoint, sheaf-theoretic cognitive system over Hilbert spaces with functorial alignment dynamics.

F Temporal Semantic Dominance Principle

Most computational and physical frameworks treat time as the primary ordering structure of reality. Events are assumed to derive meaning through temporal sequence, causality, and chronological progression. BLOC Theory reverses this assumption.

The framework proposes that semantic structure is ontologically prior to temporal ordering because cognition does not merely observe temporal flow; cognition interprets, restructures, and recursively reassigns significance to temporal states. Thus, time functions as a contextual indexing mechanism, whereas meaning functions as a higher-order relational architecture over cognitive state space.

F.1 Intuitive Interpretation

In simpler terms, the framework distinguishes between time and meaning as two fundamentally different cognitive structures. Time functions as a system for ordering events. It specifies when something happens and places experiences into a linear sequence (before, during, and after). It does not itself explain significance; it only records progression. Meaning, by contrast, functions as a system of interpretation. It assigns significance, relevance, and structure to events across time. Meaning allows an agent to reinterpret past experiences, where the same event can acquire different significance depending on later knowledge or context.

For example, consider a student who fails an examination at age 18. In temporal terms, the event is fixed: it occurred at a specific point in time with a recorded score. However, the meaning of this event is not fixed. At age 18, the student may interpret the failure as personal inadequacy. At age 30, the same individual may reinterpret the event as a formative step that redirected their career path. The temporal structure of the event does not change, but its meaning evolves through reinterpretation. This demonstrates that time preserves sequence, while meaning preserves and transforms significance.

A second example is a wedding ring. In physical and temporal terms, it is simply a material object existing in space and time. However, its meaning includes concepts such as commitment, emotional attachment, memory, and identity. These semantic properties are not contained in time itself but are assigned through cognitive interpretation. Within this framework, time is therefore treated as an indexing mechanism that organizes events, whereas meaning is treated as a higher-order relational structure that assigns interpretation over those indexed events.Consequently, cognition is modeled as operating first through meaning (interpretation) and only secondarily through temporal ordering (sequence representation).

F.2 Temporal Operators versus Semantic Operators

Let

$$\mathcal{S}_M$$

denote the semantic Hilbert space of cognitive representations.

Define the temporal evolution operator

$$T_t : \mathcal{S}_M \to \mathcal{S}_M$$

as the chronological state-transition mapping.

Define the semantic relational operator

$$\mathcal{M} : \mathcal{S}_M \times \mathcal{S}_M \to \mathbb{R}$$

which assigns semantic significance between cognitive states.

Temporal evolution produces chronological sequence:

$$x_t \mapsto x_{t+1}.$$

Semantic structure produces interpretation:

$$\mathcal{M}(x_i, x_j).$$

Unlike temporal operators, semantic operators may recursively reinterpret prior states:

$$\mathcal{M}(x_t, x_{t-k})$$

without altering chronological order itself.

Therefore semantic structure is not constrained by local temporal directionality.

F.3 Recursive Reinterpretation Principle

Human cognition continuously revises the semantic interpretation of past events while preserving their temporal position.

Thus:

$$\text{time preserves ordering,}$$

while

$$\text{meaning preserves significance.}$$

Only semantic structure possesses recursive reinterpretation capability.

Hence semantic cognition strictly dominates chronological indexing in expressive power.

F.4 Temporal Semantic Dominance Theorem

Theorem F.1 (Temporal Semantic Dominance)**.** Let

$$T_t$$

be a temporal ordering operator over

$$\mathcal{S}_M,$$

and let

$$\mathcal{M}$$

be a recursive semantic relational operator.

Assume:

1. T_t preserves chronological ordering only,
2. $\mathcal{M}$ recursively reinterprets prior semantic states,
3. semantic reinterpretation modifies cognitive significance without modifying chronology.

Then semantic relational structure possesses strictly greater representational expressivity than temporal ordering alone.

Consequently,

$$\mathcal{M} \succ T_t.$$

Proof. Temporal operators preserve state ordering:

$$x_t \mapsto x_{t+1}.$$

However, temporal evolution alone cannot modify the semantic interpretation of previously realized states.

Semantic operators instead admit recursive evaluation:

$$\mathcal{M}(x_t, x_{t-k}),$$

which allows reinterpretation of prior states while preserving temporal indexing.

Therefore semantic operators possess expressive capabilities unavailable to purely chronological operators.

Hence semantic relational structure strictly dominates temporal ordering in representational capacity. □

F.5 Consequences for Human–AI Systems

AI alignment systems operating only on temporally ordered symbolic representations cannot fully reconstruct Human semantic cognition because Human meaning operates recursively across temporal layers.

This implies:

$$\text{semantic cognition} \not\subseteq \text{temporal computation.}$$

Therefore alignment systems necessarily experience irreducible semantic compression under finite computational representation.

F.6 Philosophical Interpretation

Under BLOC Theory:

- Time is sequential.
- Meaning is recursive.
- Time indexes states.
- Meaning structures states.
- Time records transformation.
- Meaning assigns significance.

Thus meaning operates as a higher-order architectural layer over temporal flow.

BLOC therefore functions not merely as a temporal cognitive model, but as a semantic-geometric framework governing the organization of interpretive structure itself.